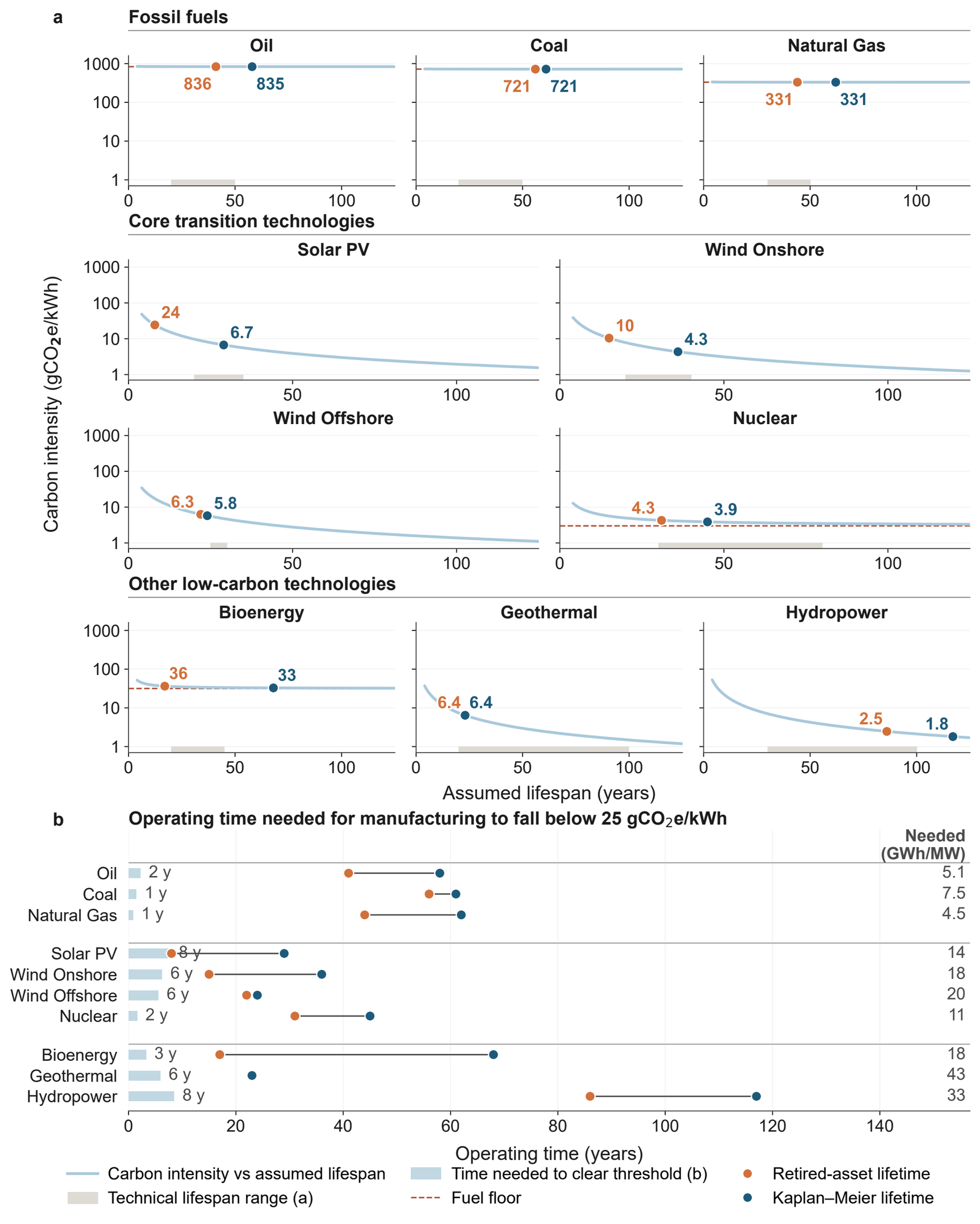

a
Fossil fuels
Oil
Coal
Natural Gas
836
835
721
721
331
331
Core transition technologies
Solar PV
Wind Onshore
24
6.7
10
4.3
Wind Offshore
Nuclear
6.3
5.8
4.3
3.9
Other low-carbon technologies
Bioenergy
Geothermal
Hydropower
36
33
6.4
6.4
2.5
1.8
Carbon intensity (gCO2e/kWh)
Assumed lifespan (years)
b
Operating time needed for manufacturing to fall below 25 gCO2e/kWh
Needed (GWh/MW)
Oil 2 y 5.1
Coal 1 y 7.5
Natural Gas 1 y 4.5
Solar PV 8 y 14
Wind Onshore 6 y 18
Wind Offshore 6 y 20
Nuclear 2 y 11
Bioenergy 3 y 18
Geothermal 6 y 43
Hydropower 8 y 33
Operating time (years)
Carbon intensity vs assumed lifespan
Technical lifespan range (a)
Time needed to clear threshold (b)
Fuel floor
Retired-asset lifetime
Kaplan–Meier lifetime